

Tristan Grote

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Dr. Ruby

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Machine Learning Model Comparison on Student Performance and Customer Purchase Behavior

This project focuses on applying and comparing multiple supervised learning models across two different datasets: Student Performance and Customer Purchase Behavior. The goal is to evaluate how different algorithms perform depending on the structure of the data and the type of prediction task. Instead of focusing on just one model, this project compares several approaches to better understand which models work best and why.

The first dataset is the Student Performance dataset, which includes demographic, social, and academic features such as study time, absences, and parental background. The original target variable (G3) represents a student's final grade. In this project, the problem was converted into a binary classification task by labeling students as passing or failing. Additionally, the earlier grades (G1 and G2) were removed to avoid data leakage, since they are highly correlated with the final grade.

The second dataset is the Customer Purchase Behavior dataset. This dataset includes features related to customer demographics and behavior, such as income, browsing activity, and previous purchases. The target variable indicates whether a customer made a purchase. This dataset was chosen because it represents a more complex, real-world classification problem involving non-linear relationships.

For both datasets, multiple supervised learning models were implemented using scikit-learn. The models used include Logistic Regression, k-Nearest Neighbors (KNN), and a Multi-Layer Perceptron (MLP) for both the student and customer datasets. Each dataset was split into training and testing sets, and feature scaling was applied where necessary, especially for models like KNN that depend on distance calculations. Hyperparameter tuning was performed using GridSearchCV to optimize model performance, particularly for KNN. These models were chosen to represent a mix of

simple linear models and more complex non-linear approaches, allowing for a meaningful comparison across different types of data.

The performance of each model varied depending on the dataset. For the Student Performance dataset, Logistic Regression achieved the best overall results. This suggests that the relationship between the input features and the binary target variable is relatively linear. K-Nearest Neighbors performed moderately well but did not outperform Logistic Regression, while the more complex models did not provide significant improvement. This indicates that adding model complexity does not always lead to better performance, especially when the underlying data structure is simple.

For the Customer Purchase dataset, the results were different. The Multi-Layer Perceptron (MLP) achieved the highest accuracy among the models tested. This suggests that the dataset contains more complex, non-linear relationships that are better captured by a neural network. Logistic regression still performed well, but it was slightly less effective than MLP, while KNN again showed reasonable performance but was not the top model. These results highlight how different datasets benefit from different types of models.

Hyperparameter tuning had mixed results. While tuning improved the performance of KNN on the Customer Purchase dataset, it did not improve performance on the Student Performance dataset. This shows that tuning does not always guarantee better results, especially if the model is already well-suited to the data. The value of k in KNN plays a key role in balancing overfitting and underfitting, and the optimal value depends on the dataset.

The Student Performance dataset has been previously studied by Cortez and Silva, who used regression methods to predict the exact final grade (G3). Their approach included features such as G1 And G2, which are highly correlated with the target variable. In contrast, this project reformulated the problem as a binary classification task and removed G1 and G2 to prevent data leakage and artificially high performance. Because of these differences, the results are not directly comparable in terms of exact values, but they highlight different modeling strategies. The prior work

focuses on precise grade prediction, while this project focuses on classification accuracy and generalization.

Overall, this project demonstrates that model performance depends heavily on the structure of the dataset and the nature of the problem. Simpler models like Logistic Regression can perform very well on structured datasets with more linear relationships, while more complex models like MLP are better suited for capturing non-linear patterns in behavioral data. One limitation of this project is the use of only two datasets, which limits how broadly the results can be generalized. In terms of real-world applications, these models could be used to identify students who are at risk of failing or to predict customer purchasing behavior for marketing purposes. The main takeaway from this project is that there is no single best model for all problems. Instead, model selection should be based on the characteristics of the data and the specific goals of the task.